MIL-F-18180/10A
30 May 1986
SUPERSEDING
MIL-F-18180/10
30 April 1981

MILITARY SPECIFICATION SHEET

FLANGES AND FLANGED FITTINGS, PIPE, STEEL, CLASSES 150, 300, 400, 600, 900, 1500, AND 2500; STRAIGHT AND REDUCING SIZE TEES

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the straight and reducing size tees described herein shall consist of this document and the latest issue of specification MIL-F-18180.



FRONT VIEW

REAR VIEW

FIGURE 1. Tees, straight and reducing size.

1. Military specification sheet part number. 90 degree, straight and reducing size, side outlet elbows covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-18180).

M18180/10	XX	X	X	X	X	X	X

Military Specification Sheet Code Number

ANSI B16.5 Material Group Number Code Number (see para 2)

"A" End Pipe Size Code Letter (see para 3)

"B" End Pipe Size Code Letter (see para 4)

"C" End (Side Outlet) Pipe Size Code Letter (see para 5)

Pressure-Temperature Rating Class Code Number (see para 6)

Flange Facing Code Letter (see para 7)

Number of Auxiliary Connection Holes (see para 8)

FSC 4730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

AMSC N/A

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

2. ANSI B16.5 material group number. The ANSI B16.5 Material Group No. for straight and reducing size, side outlet elbows is designated by two digits (see table I).

TABLE I. Code number to material group number.

Code Number	ANSI B16.5 Material Group No.	ANSI B16.5 Acceptable Material (ASTM Standards for)
01	Group 1.1	A105, A216 Grade WCB, A350 Grade LF2, A515 Grade 70, A516 Grade 70 or A537 Class 1.
02	Group 1.2	A203 Class B or E, A216 Grade WCC, A350 Grade LF3 or A352 Grade LC2 or LC3.
04	Group 1.4	A350 Grade LF1, A515 Grade 60 or A516 Grade 60.
05	Group 1.5	A182 Grade F1, A204 Grade A or B, A217 Grade WC1 or A352 Grade LC1.
07	Group 1.7	A182 Grade F2, A204 Grade C or A217 Grade WC 4 or WC 5.
09	Group 1.9	A182 Grade F11 or F12, A217 Grade WC6 or A387 Grade 11.
10	Group 1.10	A182 Grade F22, A217 Grade WC9 or A387 Grade 22.
13	Group 1.13	A182 Grade F5a or A217 Grade C5.
14	Group 1.14	A182 Grade F9 or A217 Grade C12
21	Group 2.1	A182 Grade F304 or F304H, A240 Type 304 or A351 Grade CF3 or CF8.
22	Group 2.2	A182 Grade F316 or F316H, A240 Type 316 or 317 or A351 Grade CF3M or CF8M.
23	Group 2.3	A182 Grade F304L or F316L or A240 Type 304L or 316L.
24	Group 2.4	A182 Grade F321 or F321H or A240 Type 321 or 321H.
25	Group 2.5	A182 Grade F347, F347H, F348 or F348H, A240 Type 347, 347H, 348 or 348H or A351 Grade CF8C.
26	Group 2.6	A240 Type 309S or A351 Grade CH8 or CH20.
27	Group 2.7	A182 Grade F310, A240 Type 310S or A351 Grade CK20.

Source: https://assist.dla.mil -- Downloaded: 202610-18T11:04Z
Check the source to verify that this is the current version before use.

3. End "A" nominal pipe size. The nominal pipe size for flanged end "A" (largest end except when both run ends are smaller than the outlet end) of straight and reducing size tees is designated by a single letter (see table II).

4. End "B" nominal pipe size. The nominal pipe size for flanged end "B" of straight and reducing size tees is designated by a single letter (see table II).

5. End "C" nominal pipe size. The nominal pipe size for the flanged end "C" (largest end when both run ends are smaller than the outlet end) of straight and reducing size tees is designated by a single letter (see table II).

TABLE II. Code letter to nominal pipe size.

Code Letter	Reduced size (nominal pipe size-inches)
A	1/2
B	3/4
C	1
D	1-1/4
E	1-1/2
F	2
G	2-1/2
H	3
J	3-1/2
K	4
M	5
N	6
P	8
R	10
S	12
T	14
U	16
W	18
X	20
Y	24

6. Pressure-temperature rating. The pressure-temperature rating class of flanged, straight and reducing size tees is designated by a single digit (see table III).

TABLE III. Code number to class.

Code Number	Class
1	150
3	300
4	400 1/
6	600
9	900
5	1500
2	2500

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

TABLE III. Code number to class.

Code Number	Class
1	150
3	300
4	400 1/
6	600
9	900
5	1500
2	2500

1/ Use only for nominal pipe size 4 inch through 24 inch; 1/2 inch through 3-1/2 inch nominal pipe size shall be rated as class 600 since in these sizes all dimensions for class 400 and class 600 are exactly the same.

7. Flange facing. The facing of the flanged ends is designated by a single letter (see table IV).

TABLE IV. Code letter to flange facing.

Code Letter	Flange Facing
A	Raised
B	Flat
C	Ring-Joint
D	Large Male
E	Large Female
F	Small Male
G	Small Female
H	Large Tongue
J	Small Tongue
K	Large Groove
M	Small Groove

8. Auxiliary connections. The number of auxiliary connection holes in the flanged fittings is identified by a single digit.

8.1 No connections. If there are no auxiliary connection holes, this digit is zero (0).

8.2 Number of connections. If there are auxiliary connection holes, this digit is the numerical value of the number of auxiliary connection holes (1 through 9).

8.2.1 Type and location of connections. If there are auxiliary connection holes, the holes are listed after the number of auxiliary connection holes digit and each is identified by a two-letter code indicating type of connection (B for butt-weld, S for socket weld and T for threaded) and location of connection (E, F, G, H, K, L, M, N, R, S, or T).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

8.2.2 Size of connections. If the auxiliary connection holes are not the standard size specified in ANSI B16.5, the size is prefixed to the two-letter type and location code (i.e., 3TE, 7/8 SH).

9. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the military specification sheet code number (M18180/10 for MIL-F-18180/10) the ANSI B16.5 material group number code number, the "A" end pipe size code letter, the "B" end pipe size code letter, the "C" end pipe size code letter, the pressure-temperature rating class code number, the flange facing code letter and the number of auxiliary connection holes as shown in paragraph 1.

Custodians:

Army - ME
Navy - YD

Review Activity:

DLA - CS

User Activities:

Navy - CG, MC

Preparing Activity:

Navy - YD

(Project 4730-0713)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer (156)
Naval Construction Battalion Center
Port Hueneme, CA 93043-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:04Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 20260-18T11:04Z
Check the source to verify that this is the currentversion before use.